

JEE ADVANCED - 5 | PAPER - 1

JEE 2022

Date
2nd January, 2022

Maximum Marks
180

Duration
3 Hours

General Instructions

- The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
- Section 1** contains **2 types** of questions [**Type A and Type B**].
Type A contains **SIX (06) Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
Type B contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
- Section 2** contains **THREE (03) Paragraphs**. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION-1 | Type A

- This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the answer. For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks : +3 If **ONLY** the correct option is chosen;
 - Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
 - Negative Marks : -1 In all other cases.

SECTION-1 | Type B

- This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
 - Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
 - Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
 - Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
 - Zero Marks : 0 If unanswered;
 - Negative Marks : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 - choosing **ONLY** (A), (B) and (D) will get +4 marks; choosing **ONLY** (A) and (D) will get +2 marks;
 - choosing **ONLY** (A) will get +1 mark;
 - choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 - choosing any other option(s) will get -2 marks.

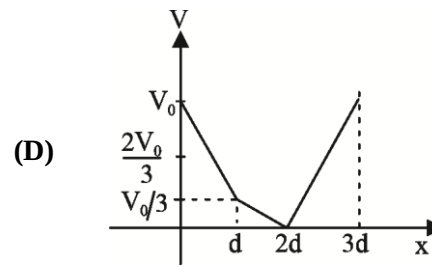
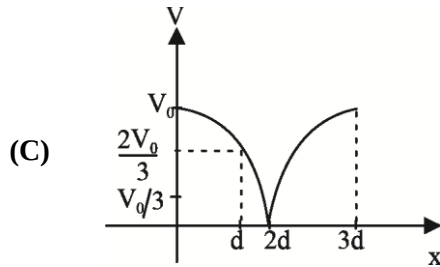
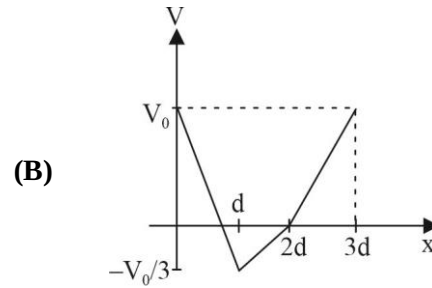
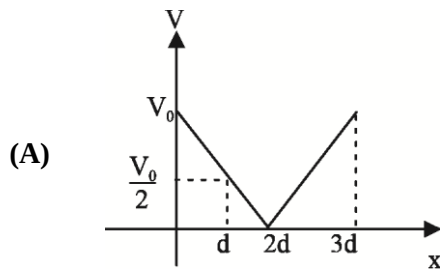
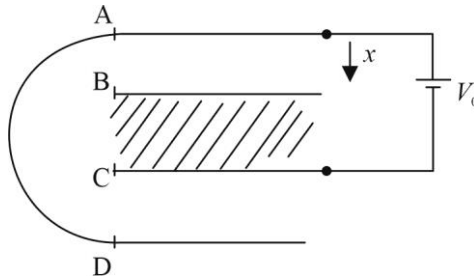
SECTION-2

- This section contains **THREE (03)** Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks:** +3 If **ONLY** the correct Integer value is entered. There is **NO negative marking**.
 - Zero Marks:** 0 In all other cases.

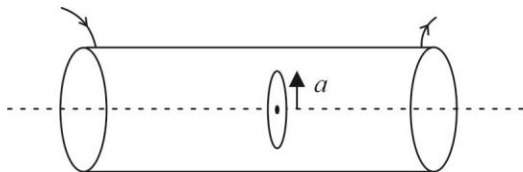
SECTION-1 | Type A

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

1. Four identical thin plates A, B, C, D are placed parallel to each other at equal distance d from each other. Space between plates B and C is filled with a dielectric of dielectric constant $k = 2$. Now plates A and C are connected across a battery of $EMF V_0$ as shown. Assuming potential of plate C as zero and ignoring the thickness of plates, variation of potential (V) vs x is best represented by (x is measured from plate A).



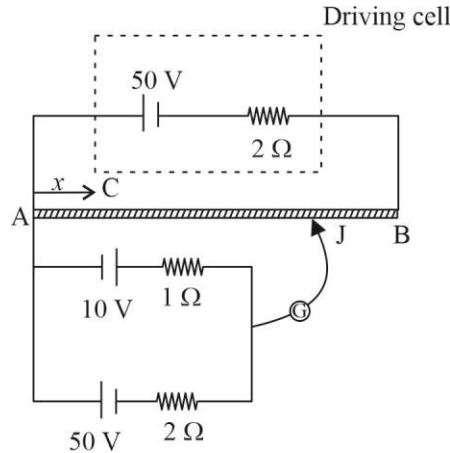
2.



A circular ring of radius a , having resistance R is placed coaxially inside a long solenoid with ' n ' number of turns per unit length. If current through the solenoid is varied according to $I = I_0 t$, what is the maximum current through the ring? [L : self inductance of Ring]

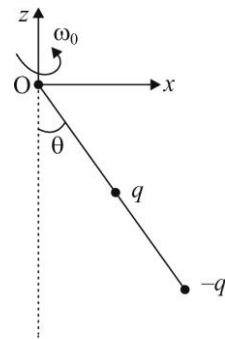
- (A) $\frac{LI_0}{R}$ (B) $\frac{\mu_0 n \pi a^2 I_0}{R}$ (C) $\frac{(\mu_0 n \pi a^2 + L) I_0}{R}$ (D) Infinite

3. In a potentiometer experiment setup shown below potentiometer wire (AB) of length 1 m is used and resistivity of potentiometer varies in such a way that resistance of length x (AC) of potentiometer wire from left end is proportional to x^2 . Resistance of the potentiometer wire is 8Ω . Null point is obtained for value of x equal to :



- (A) 0.37 m (B) 0.5 m (C) 0.54 m (D) 0.75 m

4. An insulating rod of length ℓ is rotating about z-axis with constant angular velocity $\omega_0 \hat{k}$. Rod makes an angle $\theta = 37^\circ$ with Z-axis and two charges q and $-q$ are attached at the centre and end of the rod. Magnetic field at O the moment rod is in x-z plane as shown is:



- (A) $\frac{3\mu_0 q \omega_0}{20\pi\ell} \left(\frac{4}{5} \hat{i} + \frac{3}{5} \hat{k} \right)$ (B) $\frac{3\mu_0 q \omega_0}{20\pi\ell} \left(\frac{3}{5} \hat{i} + \frac{4}{5} \hat{k} \right)$
 (C) $\frac{\mu_0 q \omega_0}{4\pi\ell} \left(\frac{3}{5} \hat{i} + \frac{4}{5} \hat{k} \right)$ (D) $\frac{\mu_0 q \omega_0}{4\pi\ell} \left(\frac{4}{5} \hat{i} + \frac{3}{5} \hat{k} \right)$
5. Two non-conducting spheres A & B of radius a and $b \left(= \frac{a}{2} \right)$ are uniformly charged with charges $-Q_0$ and $2Q_0$ respectively. Separation between the centre of two spheres is $2a$. If $\vec{E}_A$ and $\vec{E}_B$ represent electric field at a point due to spheres A and B respectively then value of $\int \epsilon_0 |\vec{E}_1 + \vec{E}_2|^2 dv$ over entire space is [dv is a small volume in space].

- (A) $\frac{16Q_0^2}{5\pi\epsilon_0 a}$ (B) $\frac{11Q_0^2}{5\pi\epsilon_0 a}$ (C) $\frac{8Q_0^2}{5\pi\epsilon_0 a}$ (D) $\frac{27Q_0^2}{20\pi\epsilon_0 a}$

6. Intensity of Electric field depends only on coordinates x and y as $E = E_0 \frac{y\hat{i} - x\hat{j}}{x^2 + y^2}$

Electric flux through a cylinder of radius R , height H with axis along z-axis is :

- (A) $4\pi\epsilon_0 E_0 R$ (B) $2\pi H E_0$ (C) $4\pi H E_0$ (D) Zero

SPACE FOR ROUGH WORK

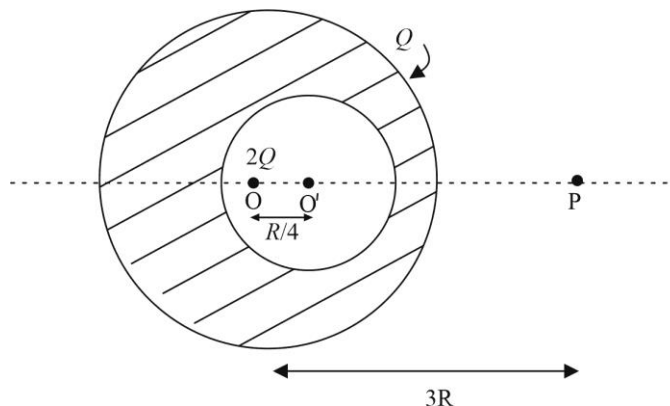
SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Consider two parallel electron beams moving in same direction. Cross sectional area of each beam is A , v is speed of electron in the beam, n is electron concentration in the either beam, r is distance between the two beams, e is electronic charge. Cross-section of the beam is small in comparison to r . Choose the correct statement(s):

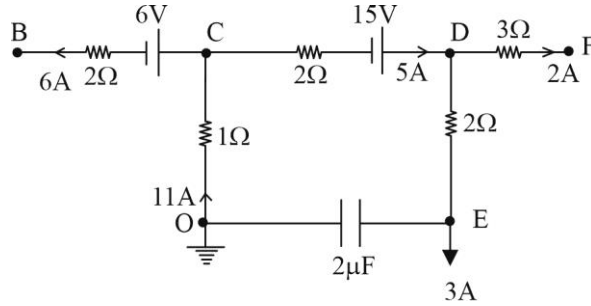
- (A) Electric force acting on an element of length l , chosen in any of the beams is $\frac{e^2 n^2 A^2 \ell}{2\pi\epsilon_0 r}$
- (B) Magnetic force acting on an element of length l is $\frac{\mu_0 e^2 n^2 v^2 A^2 l^2}{2\pi r}$
- (C) Ratio of magnetic force to electric force acting on an element of length l in any of the beams is $\frac{v^2}{c^2}$
- (D) Beams will eventually converge due to attractive magnetic force.

8. A conducting sphere of radius R has a spherical cavity of radius $\frac{R}{2}$ in it. O is the centre of sphere and O' is the centre of cavity. The distance OO' is $\frac{R}{4}$. A charge Q is given to the sphere and another point charge $2Q$ is placed at O . Choose the correct statements.

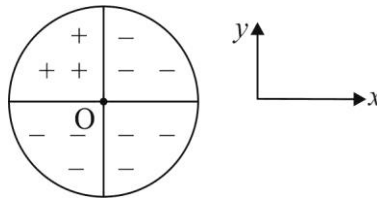


- (A) Electric field at a distance $3R$ from the center of sphere will be $\frac{1}{4\pi\epsilon_0} \frac{Q}{9R^2}$
- (B) Potential at point O' is $\frac{7Q}{4\pi\epsilon_0 R}$
- (C) Field at a point P outside the sphere due to charges on the inner surface of the cavity will be $\frac{Q}{18\pi\epsilon_0 R^2}$
- (D) If another charge q is brought from infinity and placed outside the sphere, potential of O' will remain unchanged

9. In the circuit below, choose the correction statements.



- (A) Potential of B is -29 V (B) Potential of D is -6V
(C) Energy stored in the capacitor is $144 \mu J$ (D) Potential of C is 11 V.
10. A conducting liquid of resistivity ρ is flowing inside a parallel plate capacitor, with constant speed v_0 . Plates of the capacitor have area A and separation between the plates is d . The capacitor plates are connected with an external resistor of resistance R . A uniform magnetic field B is applied parallel to the plates and normal to flow velocity of liquid. choose the correction statements. [Ignore end effects]
- (A) Surface charge density on the upper surface of liquid in contact with plates of capacitor is $\frac{Bv_0\epsilon_0}{2}$
(B) Surface charge density on the upper surface of liquid in contact with plates of capacitor is $Bv_0\epsilon_0$
(C) Current through resistance R is $\frac{Bv_0d}{R}$
(D) Maximum power dissipated in R can be $\frac{B^2v_0^2Ad}{4\rho}$
11. A non-conducting circular plate of radius R has uniform surface charge density $-\sigma$ on three quarters and $+\sigma$ on one of the quarters as shown below. Disc lies in x - y plane with its centre at origin. Choose the correct statements.



- (A) Electric field at a point $(0, y, 0)$ on y -axis points in $+x$ direction ($y > R$)
(B) Electric field at a point on line $y + x = 0$ in x - y plane is along $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$
(C) E_z , z -component of electric field at a point $(0, 0, R)$ is $\frac{-\sigma}{4\epsilon_0} \left[1 - \frac{1}{\sqrt{2}} \right]$
(D) Work done by electric field in moving a charge q from $(0, 0, R)$ to $(0, 0, 2R)$ is $\frac{Rq\sigma}{4\epsilon_0} [\sqrt{5} - \sqrt{2} - 1]$

12. Temperature coefficient of resistivity (δ) is given as $\alpha = \frac{1}{\delta} \frac{d\delta}{dT}$ where δ is resistivity of material at temperature T . Choose the correct option(s) :
- (A) $\delta = \delta_0 e^{\alpha(T-T_0)}$; δ_0 is resistivity at temperature T_0 .
- (B) $\delta = \delta_0 (1 + \alpha(T - T_0))$ if $\alpha(T - T_0) \ll 1$.
- (C) Resistance of a cylindrical wire at temperature T is given by

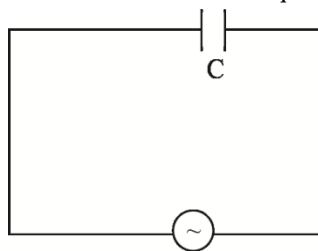
$$R = \frac{R_0 [1 + \alpha(T - T_0)] [1 + \alpha'(T - T_0)]}{1 + 2\alpha'(T - T_0)}$$
, where R_0 is resistance of wire at temp T_0 and α' is coefficient of linear expansion and α is temperature coefficient of resistivity
- (D) With increase in temperature, number density of conduction electrons and average relaxation time decrease in a conductor and so its resistivity increases.

SECTION-2

This section consists of THREE (03) Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

An AC source with voltage $V = 4 + 3\sin(2\pi t)$ (V is in volts and t in seconds) is connected across a capacitor with capacitance $C = 1F$. Given AC source can be seen as combination of two sources $V_1 = 4V$ and $V_2 = 3\sin(2\pi t)$ in series and in order to determine current through an element connected across it principle of superposition may be applied. In this principle current through the element is found by taking one source at a time and switching off the other and then superimposing the two currents. It may be noted that principle of superposition will not work for non linear quantities like power.



- What is the magnitude of average power (in watt) delivered to the capacitor in $t = 0.5\text{ s}$ to 0.75 s ?
- If a resistor with resistance R ($2\sqrt{3}\pi RC = 1$) is connected in series with capacitor, the current in it is found to be $\frac{A}{R} + \frac{B}{4R} \sin(2\pi t + \phi)$, value of $A + B$ is _____.

Paragraph for Question 3-4

A charged particle having charge 4 mC and mass 4g enters a region of uniform magnetic field.

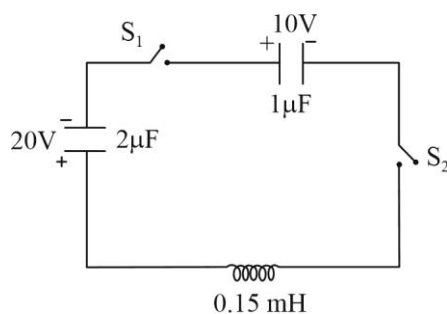
$\vec{B} = -\frac{\sqrt{65}}{6} \hat{k}$ (Tesla) and an electric field $\vec{E} = (2x+2)\hat{i}$ N/C where x is in meters. Charge enters this

region from origin with velocity $\vec{v} = 3\hat{i}$ m/s.

3. What is radius of curvature (in meters) of the trajectory just after entering the region?
4. Find speed (in m/s) of the particle when its x -coordinate is $x = 2$.

Paragraph for Question 5-6

Two capacitors of capacitance $1\mu F$ and $2\mu F$ are charged to potential difference 10 V and 20 V respectively and connected to an inductor of inductance $L = 0.15$ mH as shown in figure. Switches S_1 and S_2 are closed at $t = 0$.

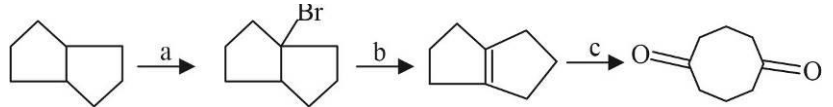


5. Potential difference across capacitor of capacitance $1\mu F$ when current in the circuit is maximum is _____ volts .
6. Maximum current in the circuit is _____ Amperes.

SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY**60 MARKS****SECTION-1 | Type A**

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- The sublimation temperature of a substance is -83°C . It exists as cubic close packed structure with unit cell edge length of $5.0 \text{ \AA}$ and density $\frac{8}{3} \text{ g / cm}^3$ at -100°C . What would be the density of substance at 0°C and 1 atm assuming $N_A = 6 \times 10^{23}$.
 (A) 1.86 g / cm^3 (B) 2.23 g / cm^3 (C) 2.23 g / litre (D) 1.86 g / litre
- If two liquids A & B form minimum boiling azeotrope at a specific composition then:-
 I. A – B interactions are stronger than A – A or B – B interactions.
 II. Vapour pressure of solution increases because of a greater number of molecules of liquids A & B can escape from the solution.
 III. Vapour pressure of solution decreases because of a smaller number of molecules of one of the liquids escape from the solution.
 IV. A – B interactions are weaker than A – A or B – B interactions.
 (A) I, III (B) II, IV (C) I only (D) II only
- In solid state, PCl_5 is a/an:-
 (A) Covalent Solid
 (B) Octahedral Structure
 (C) Ionic solid with $[\text{PCl}_6]^+$ Octahedral Structure and $[\text{PCl}_4]^-$ Tetrahedral Structure
 (D) Ionic solid with $[\text{PCl}_4]^+$ Tetrahedral Structure and $[\text{PCl}_6]^-$ Octahedral Structure
- What will be a, b and c respectively in the below mentioned conversion?

 (A) $\text{Br}_2 / h\nu$; $\text{MeO}^- / \text{MeOH}$; $\text{O}_3 / \text{Me}_2\text{S}$
 (B) $\text{NBS} / \text{CCl}_4$; NaNH_2 ; $\text{O}_3 / \text{H}_2\text{O}$
 (C) $\text{Br}_2 / h\nu$; $\text{EtO}^- / \text{EtOH}$; $\text{KMnO}_4 / \text{OH}^- / 0^{\circ}\text{C}$
 (D) Both A or C
- Select the incorrect statement
 (A) Sucrose is a non reducing sugar and comprises $\text{C}_1 - \text{C}_2$ glycosidic linkage
 (B) Maltose is a reducing sugar and comprises $\text{C}_1 - \text{C}_4$ glycosidic linkage
 (C) Lactose is a reducing sugar and upon hydrolysis gives two units of α -D-glucose
 (D) Fructose gives positive tollen's test

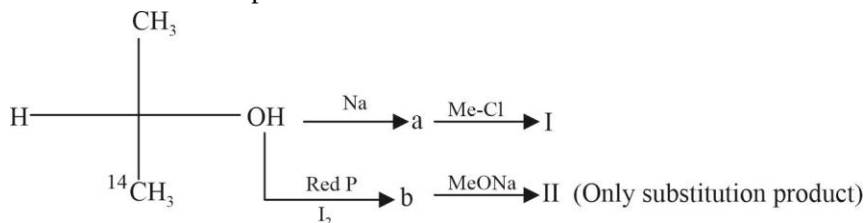
6. α – Helix is a secondary structure of proteins formed by twisting of polypeptide chain into right-handed screw like structures. Which type of interactions are responsible for making the α -helix structure stable?
- (A) London-Dispersion forces (B) Ionic bond
(C) H-bonding (D) Disulphide linkages

SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Consider 0.1 m aqueous solution of two solute X and Y. The solute X behaves as a weakly ionizing electrolyte, while Y dimerises in solution. Which of the following statement(s) is/are correct regarding these solutions?
- (A) The boiling point of the solution of X will be higher than that of Y
(B) The osmotic pressure of the solution of X will be higher than that of Y
(C) The freezing point of the solution of X will be higher than that of Y
(D) The relative lowering of vapour pressure of both the solution will be the same
8. KCl crystals appear violet or lilac in colour. Choose the correct statement(s)
- (A) Some of the anionic sites are occupied by an unpaired electron.
(B) Some of the anionic sites are occupied by a pair of electrons.
(C) KCl crystals have metal excess defect
(D) F-centres are created which impart colour to the crystals.
9. Which of the following statements are correct?
- (A) S–S bond is present in $\text{H}_2\text{S}_2\text{O}_6$
(B) In H_2SO_5 sulphur is in +8 oxidation state.
(C) Preparation of NH_3 by Haber's process is endothermic in nature. Hence high temperature is used to shift the equilibrium towards ammonia.
(D) Change in enthalpy is negative for the preparation of SO_3 by catalytic oxidation of SO_2
10. A compound 'H' ($\text{C}_8\text{H}_9\text{BrO}$) is obtained on treatment of an aromatic chiral compound 'A' ($\text{C}_8\text{H}_8\text{Br}_2$) with aqueous NaOH. On heating 'A' with potassium tert-butoxide, 'I' ($\text{C}_8\text{H}_7\text{Br}$) is formed. 'A' on reaction with one equivalent of methyl magnesium bromide in ether, p-bromoisopropyl benzene is formed. Which observations are correct about these reactions?
- (A) 'H' is an optically active phenol
(B) 'H' gives violet coloration with neutral FeCl_3 solution.
(C) 'H' gives positive iodoform test
(D) 'I' is optically inactive and doesn't exhibit geometrical isomerism

11. Choose the correct option



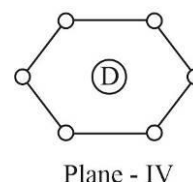
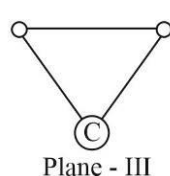
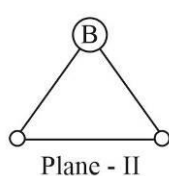
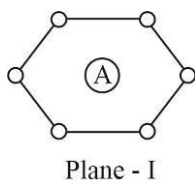
- (A) I and II are identical
 (B) I and II are different
 (C) Conversion of 'a' to I, and 'b' to II obey S_N1 mechanism
 (D) Conversion of 'a' to I, and 'b' to II obey S_N2 mechanism
12. In a AB unit cell (Rock salt type) which of the following statement is / are correct :
- (A) The nearest neighbours of A^+ are 6
 (B) The nearest neighbours of B^- are 6
 (C) The second nearest neighbours of A^+ are 12
 (D) The nearest distance between two A^+ ions is $\frac{r_A + r_B}{\sqrt{2}}$, where r_A and r_B are the radius of cation and anion respectively.

SECTION-2

This section consists of THREE (03) Paragraphs. There are **TWO (02)** questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

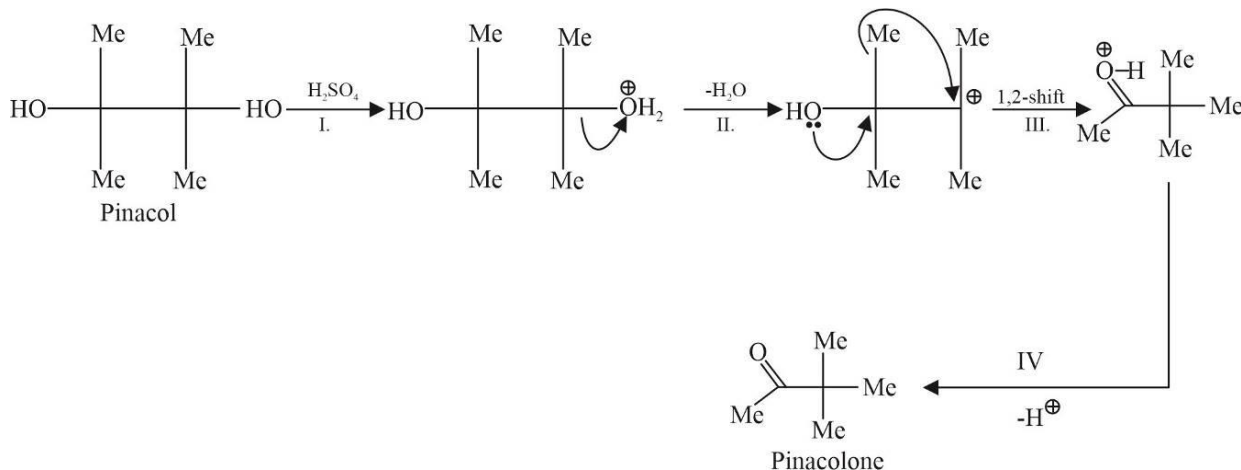
In a box two thousand perfectly spherical shaped hard tennis balls are closely packed. The balls took the following array. The radius of each ball is 2 cm



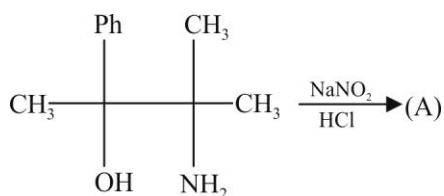
1. The effective number of atoms in above packing per unit cell is :
2. If the distance between the centers of the balls A and D is ℓ (cm) then the value of $\frac{\ell}{\sqrt{6}}$ is :

Paragraph for Question 3-4

Given below is the mechanism of Pinacol-Pinacolone rearrangement. Observe the mechanism carefully and answer the following question.



3.



Number of C – N bonds in the product A is :

4. Product A reacts with methylamine to give B. 'B' further reacts with NaBH_3CN to give 'C'. If number of possible stereoisomers of 'B' = x and if degree of unsaturation of 'C' = y. Then find value of (x + y) ?

Paragraph for Question 5-6

An organic compound ($\text{C}_x\text{H}_{2y}\text{O}_y$) was burnt with twice the amount of oxygen needed for complete combustion to CO_2 and H_2O . The hot gases when cooled to 0°C and 1 atm pressure, measured 2.24 L. The water collected during cooling weigh 0.9 g. The vapour pressure of pure water at 20°C is 50.3 mm Hg and is lowered by 0.3 mm when 50 g of the organic compound is dissolved in 1000 g of water.

5. The value of $\frac{x+y}{2} + 1$ is.
6. If the molecular mass of the given organic compound is M then the value of $\frac{M}{10} - 9$ is.

SPACE FOR ROUGH WORK

SUBJECT III : MATHEMATICS**60 MARKS****SECTION-1 | Type A**

This section consists of 6 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- Suppose $f : R \rightarrow R$ be a continuous function such that $f(3x) = f(x) \forall x \in R$ and $f(2) = 5$, then value of $\int_2^5 \operatorname{sgn}(f(x)) dx$ is equal to :
 (A) 0 (B) 1 (C) -3 (D) 3
- Let $f : [0, \infty) \rightarrow R$ be a continuous and strictly increasing function such that $f^4(x) = \int_0^x t^2 f^3(t) dt \forall x > 0$. The area enclosed by $y = f(x)$, the x-axis and the ordinates at $x = -1$ and $x = 4$ is :
 (A) $\frac{17}{48}$ (B) $\frac{65}{48}$ (C) $\frac{257}{48}$ (D) $\frac{255}{48}$
- The sum $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{2n+1}{n^4 + 2n^3 + n^2 + 1}\right)$ is equal to :
 (A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$
- If $P(r)$ be number of points (x, y) such that $x, y \in I$ which lie inside or on the curve $|xy| - r(|x| + |y|) + r^2 = 0, r \in N$ & if $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^{n-1} \sqrt{P(r)}}{an^3 + bn + c} = \frac{1}{2}$ then $a+b$ is :
 (A) 2 (B) 3 (C) 4 (D) 1
- Let $g(x)$ be a polynomial of degree one and $f(x)$ be defined by $f(x) = \begin{cases} g(x) & x \leq 0 \\ |x|^{\sin x} & x > 0 \end{cases}$. If $f(x)$ is continuous satisfying $f'(1) = f(-1)$, then find $g(-101)$.
 (A) $101\sin 1 - 100$ (B) $101\sin 1 + 100$ (C) $100\sin 1 - 101$ (D) $100\sin 1 + 101$
- Consider the function $f(x) = \left(1 + \frac{1}{x}\right)^x$ then find correct statement ?
 (A) $f(x)$ has a maxima but no minima
 (B) $f(x)$ has a minima but no maxima
 (C) $f(x)$ has exactly one maxima and one minima
 (D) has neither a maxima nor a minima

SECTION-1 | Type B

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Consider $f(x) = \frac{\cos^{-1}(\cos(\sin x)) - |x - \pi|}{\sin^3 x}$, then :
- (A) $\lim_{x \rightarrow 0} f(x)$ does not exist (B) $f(x)$ has removable discontinuity at $x = \pi$
- (C) Jump of discontinuity at $x = \pi$ is $\frac{1}{3}$ (D) $f(x)$ is discontinuous at all $x = n\pi, n \in I$
8. If $f(x) + f(y) = \frac{1}{x} + \frac{1}{y} \forall x, y \in R - \{0\}$ and $\int_2^3 \frac{3(f(x))^5 - f(x)}{1 - (f(x))^4} dx = \frac{1}{2} \log \frac{2^\alpha}{3^\beta}$, then :
- (A) $\alpha > \beta$ (B) β is prime (C) $\alpha < 2\beta$ (D) $(\alpha + \beta)$ is prime
9. Which of the following is true ?
- (A) $\frac{1}{2} < \int_0^{0.5} \frac{dx}{\sqrt{1-x^{2n}}} < \frac{\pi}{6} \quad n \in N$
- (B) The values of a and b for which the function $y = a \ln x + bx^2 + x$ has extrema at $x_1 = 1$ & $x_2 = 2$ are $a = \frac{-2}{3}, b = \frac{-1}{6}$
- (C) $y = \cos^2 x + \cos^2 \left(x + \frac{\pi}{3}\right) - \cos x \cos \left(\frac{\pi}{3} + x\right)$ then $\frac{dy}{dx} = 0$
- (D) $f(x) = \int_0^x \sqrt{1-t^4} dt$ is odd function
10. The function $f(x) = x^{1/3}(x-1)$.
- (A) has vertical tangent at $x = 0$
- (B) area enclosed by curve with x-axis from $x = -1$ to $x = 1$ is $\frac{3}{2}$
- (C) is concave up in $(0, \infty)$
- (D) is decreasing for $x \in \left(-\infty, \frac{1}{4}\right)$
11. If $f(x) = \int_0^x |t-1| dt$, where $0 \leq x \leq 2$, then find the correct statements.
- (A) range of $f(x)$ is $[0, 1]$ (B) $f(x)$ is differentiable at $x = 1$
- (C) $f'(x) = \cos^{-1} x$ has two real roots (D) $f'(1/2) = 1/2$

12. Let $f : I - \{-1, 0, 1\} \rightarrow [-\pi, \pi]$ be defined as $f(x) = 2 \tan^{-1} x - \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ then which of the following statement(s) is(are) correct ?
- (A) $f(x)$ is bijective (B) $f(x)$ is injective but not surjective
 (C) $f(x)$ is neither injective nor surjective (D) $f(x)$ is an odd function
- [Note : I denotes the set of integers]

SECTION-2

This section consists of THREE (03) Paragraphs. There are TWO (02) questions corresponding to each Paragraph. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

Paragraph for Question 1-2

For all $x \in \left[-\frac{7}{2}, \frac{7}{2} \right]$, let $f(x) = \begin{cases} [x] & , 0 \leq \{x\} < \frac{1}{2} \\ |x| & , \frac{1}{2} \leq \{x\} < 1 \end{cases}$.

[Note : $[x]$ denotes greatest integer less than or equal to x and $\{x\}$ denotes fractional part of x]

- Number of solution(s) of the equation $f(x) = \frac{5}{2}$ is equal to :
- If L is the number of points of discontinuity and M is the number of points of non-differentiability of the function $f(x)$, then $(L + M)$ is equal to :

Paragraph for Question 3-4

Consider a polynomial for $f(x) = a^2x^3 + ax^2 - x + b$ with local minimum at $x = -1$ with positive extremum values.

- Let $S_1 = \{b \mid b \in I \text{ for } a \in I\}$ and $S_2 = \{b \mid b \in I \text{ for } a \notin I\}$.

Then maximum value of $S = \sum_{n \in S_1} \left(\frac{\cos \theta}{2} \right)^n + \sum_{p \in S_2} \left(\frac{\cos(\theta/2)}{3} \right)^p$.

- If $a = 1$, $b = 2$ and $g(x) = \begin{cases} \max(f(t)) & t \leq x \quad x \leq 0 \\ \min(f(t)) & 0 \leq t \leq x \quad x > 0 \end{cases}$, then value of $\frac{1}{3}g(0) + 27g(5)$.

Paragraph for Question 5-6

Let f be cubic polynomial function s.t
$$\begin{bmatrix} a^2 & 4a & 1 \\ b^2 & 4b & 1 \\ c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) & 0 \\ f(1) & f(0) \\ f(2) & 1 \end{bmatrix} = \begin{bmatrix} 2a^2 + 16a + 17 & 4a + 1 \\ 2b^2 + 16b + 17 & 4b + 1 \\ 2c^2 + 16c + 17 & 4c + 1 \end{bmatrix}.$$

5. Number of real roots of $f(x) = 0$.
6. The sum of greatest and least values of $f(x)$ in $[-1, 1]$.

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